



**Probability and  
Statistics -  
Practice Tests  
and Solutions**



# 1. Probability



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# Definition:

## Classic Model

Number of outcomes in which  
the event occurs

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Total Number of possible  
outcomes of an experiment

## Relative Frequency of Occurrence

Number of times an event  
occurred

---

Total number of opportunities for  
an event to occur

*Probability*

# Rule of Addition

The probability that Event A or Event B occurs

=

Probability that Event A occurs

+

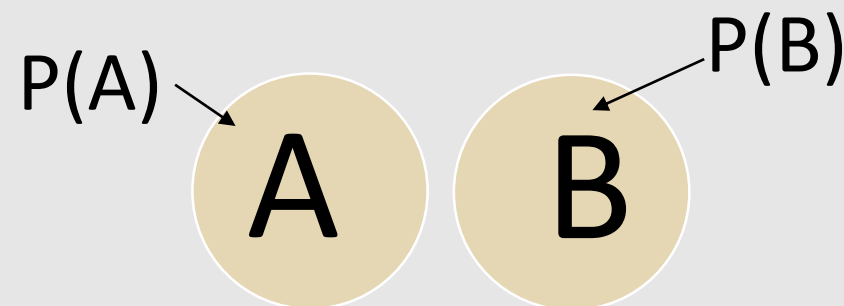
Probability that Event B occurs

-

Probability that both Events A and B occur

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

**Mutually Exclusive Events:** When two events cannot occur at the same time  $P(A \cap B) = 0$

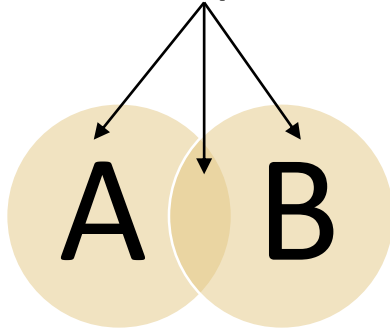


**Independent Events:** The occurrence of Event A does not change the probability of Event B

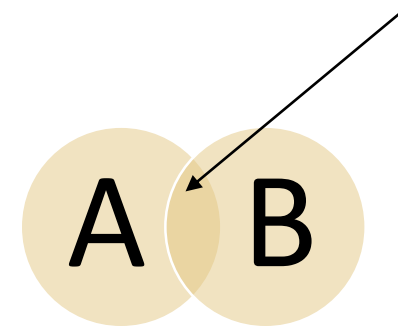
$$P(A \cap B) = P(A) P(B|A)$$

# Venn Diagram

❖ Union: Probability that events A or B occur:  $P(A \cup B)$



❖ Intersection: Probability that events A and B occur:  $P(A \cap B)$



|                                                  | With repetition             | Without repetition              |
|--------------------------------------------------|-----------------------------|---------------------------------|
| <b>❖ Permutation:</b><br>Order is important.     | $n^r$                       | $n_{Pr} = \frac{n!}{(n-r)!}$    |
| MS Excel Formula >>                              | PERMUTATIONA(n,r)           | PERMUT(n,r)                     |
| <b>❖ Combination:</b><br>Order is NOT important. | $\frac{(r+n-1)!}{r!(n-1)!}$ | $n_{Cr} = \frac{n!}{(n-r)! r!}$ |
| MS Excel Formula >>                              | COMBINA(n,r)                | COMBIN(n,r)                     |

# PERMUTATIONS / COMBINATIONS SUMMARY

# Bayes' Theorem

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|B') \cdot P(B')}$$

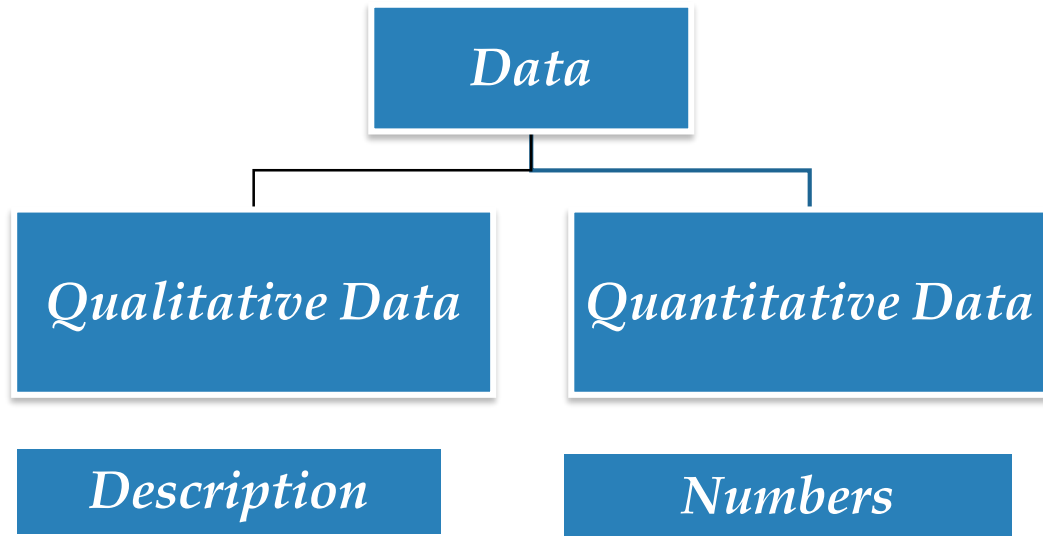


## 2. Descriptive Statistics



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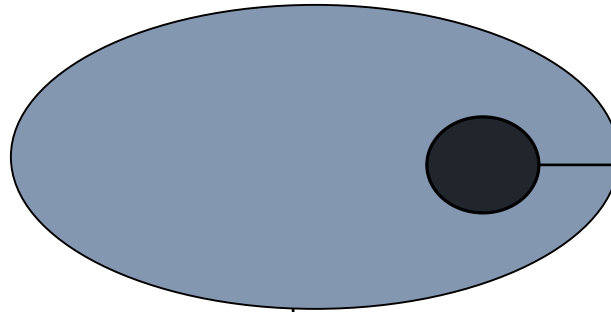




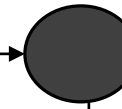
- ❖ Continuous Data
  - ❖ Measurements: Length, height, time
- ❖ Discrete Data
  - ❖ Count: Number of students, Number of heads

# Data

Population:  
Complete  
collection to  
be studied



Sampling  
Process



Sample: Part  
of population

$\mu$  – population mean  
 $\sigma$  – population std. dev.



Characteristic of  
a population

Inference



Characteristic  
of a sample

$\bar{x}$  – sample mean  
 $s$  – sample std. dev.

# SAMPLING

# Mean

$$\mu = \frac{\sum x}{N}$$

# Median

Middle value when put in ascending or descending order.

# Mode

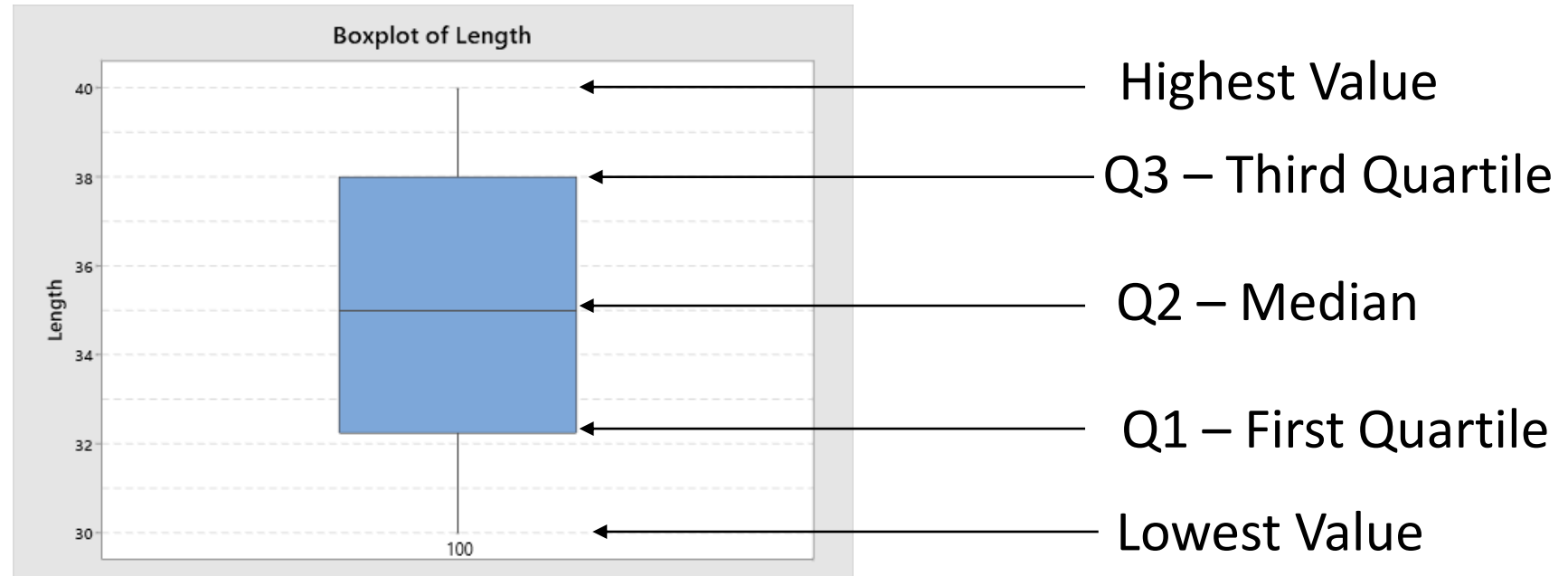
Most occurring value

# Standard Deviation

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}} \quad s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Variance is the square of the standard deviation.

## Descriptive Statistics



# Box and Whisker Plot



## 3A Binomial Distribution



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**A binomial experiment** has the following properties:

- ❖ The experiment consists of  $n$  repeated trials.
- ❖ Each trial can result in just two possible outcomes. We call one of these outcomes a success and the other, a failure.
- ❖ The probability of success, denoted by  $p$ , is the same on every trial.
- ❖ The trials are independent; that is, the outcome on one trial does not affect the outcome on other trials.

$$P(x) = {}_nC_x \cdot p^x \cdot (1 - p)^{n - x}$$



- 
- $n$ : number of trials
  - $x$ : The number of successes desired
  - $p$ : The probability of getting success in one trial

# **BINOMIAL DISTRIBUTION**

- ❖ The mean of the distribution ( $\mu_x$ ) is

$$n \cdot p$$

- ❖ The variance ( $\sigma^2_x$ ) is

$$n \cdot p \cdot (1 - p)$$

- ❖ The standard deviation ( $\sigma_x$ ) is

$$\sqrt{n \cdot p \cdot (1 - p)}$$

# BINOMIAL DISTRIBUTION



## 3B Poisson Distribution



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A **Poisson experiment** has the following properties:

- ❖ The experiment results in outcomes that can be classified as successes or failures.
- ❖ The average number of successes ( $\mu$ ) that occurs in a specified region is known.
- ❖ Outcomes are random. Occurrence of one outcome does not influence the chance of another outcome of interest.
- ❖ The outcomes of interest are rare relative to the possible outcomes.
  - ❖ Example: Road accidents, que at the counter

$$P(x, \mu) = e^{-\mu} \cdot \frac{\mu^x}{x!}$$

- $e$ : A constant equal to approximately 2.71828. (Actually,  $e$  is the base of the natural logarithm system)
- $\mu$ : The mean number of successes that occur in a specified region.
- $x$ : The actual number of successes that occur in a specified region.
- $P(x; \mu)$ : The **Poisson probability** that exactly  $x$  successes occur in a Poisson experiment, when the mean number of successes is  $\mu$ .

# POISSON DISTRIBUTION

- The Poisson distribution has the following properties:
- The mean of the distribution is equal to  $\mu$  .
- The variance is also equal to  $\mu$  .

## **BINOMIAL DISTRIBUTION**



## 3C Normal Distribution



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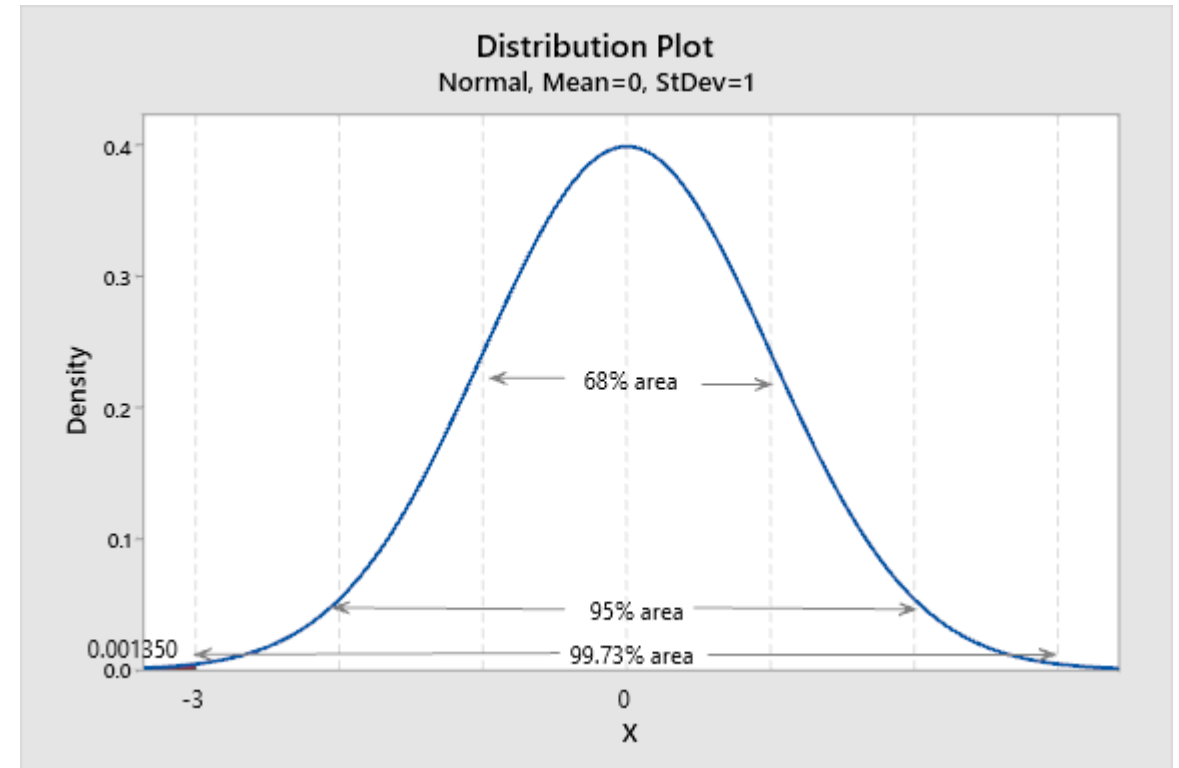
$$P(x) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}$$

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- where  $x$  is a normal random variable,
- $\mu$  = mean,
- $\sigma$  = standard deviation,
- $\pi$  is approximately 3.14159,
- $e$  is approximately 2.71828.

# NORMAL DISTRIBUTION

- About 68% of the area under the curve falls within **1 standard deviation** of the mean.
- About 95% of the area under the curve falls within **2 standard deviations** of the mean.
- About 99.7% of the area under the curve falls within **3 standard deviations** of the mean.



# NORMAL DISTRIBUTION

## Z Value / Standard Score

- ❖ How many standard deviations an element is from the mean.
- ❖  $z = (x - \mu) / \sigma$
- ❖  $z$  is the z-score,
- ❖  $x$  is the value of the element,
- ❖  $\mu$  is the population mean,
- ❖  $\sigma$  is the standard deviation.

# NORMAL DISTRIBUTION

# Binomial Distribution approximated to the Normal Distribution

If  $np$  and  $n(1-p) \geq 10$  then approximate it to The Normal Distribution

The mean of the distribution ( $\mu_x$ ) is

$$n \cdot p$$

The standard deviation ( $\sigma_x$ ) is

$$\sqrt{n \cdot p \cdot (1 - p)}$$

## NORMAL DISTRIBUTION



## 3D t-Distribution

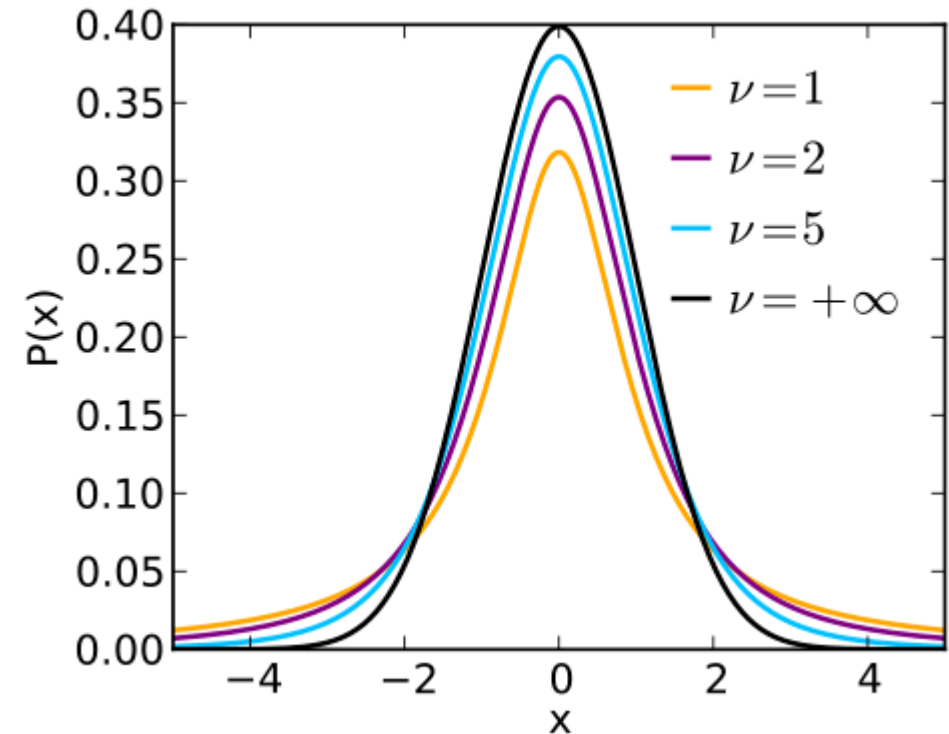


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$$t = \frac{(\bar{x} - \mu)}{s/\sqrt{n}}$$

- ❖  $s$  is the sample standard deviation
- ❖  $\bar{x}$  is the sample mean
- ❖  $n$  is the number of samples
- ❖  $\mu$  is the population mean



Reference: Wikipedia

## t - DISTRIBUTION



## 4 Hypothesis Testing Basics



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❖ For almost all populations, the sampling distribution of the mean can be approximated closely by a normal distribution, provided the sample size is sufficiently large.

❖ If a variable has a mean of  $\mu$  and the variance  $\sigma^2$ , as the sample size  $n$  increases, the sample mean approaches a normal distribution with mean  $\mu_{\bar{x}}$  and variance  $\sigma_{\bar{x}}^2$

$$\mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{n}$$

The standard deviation of the sampling distribution of the mean is called the standard error of the mean.

$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}$$

# CENTRAL LIMIT THEOREM

### ❖ Point estimate:

- ❖ Summarize the sample by a single number that is an estimate of the population parameter.
- ❖ The sample mean  $\bar{x}$  is a point estimate of the population mean  $\mu$ . The sample proportion  $\hat{p}$  is a point estimate of the population proportion  $P$ .

### ❖ Interval estimate:

- ❖ A range of values within which, we believe, the true parameter lies with high probability.
- ❖ For example,  $a < \bar{x} < b$  is an interval estimate of the population mean  $\mu$ . It indicates that the population mean is greater than  $a$  but less than  $b$ .

# CONFIDENCE INTERVAL

$$CI = \bar{x} \pm \frac{(z_{\alpha/2}) \cdot \sigma}{\sqrt{n}}$$

- ❖  $z_{\alpha/2}$  = z table value for confidence level,
- ❖  $\sigma$  = standard deviation
- ❖  $n$  = sample size.

$$CI = \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- $z_{\alpha/2}$  = z table value for confidence level
- $\hat{p}$  = sample proportion
- $n$  = sample size

# CONFIDENCE INTERVAL

## TIPS

- ❖ Null Hypothesis: The person is innocent
- ❖ Alternate Hypothesis: The person is guilty. You need to provide proof of this.
- ❖ In statistical terms you:
  - ❖ Reject the Null Hypothesis, or
  - ❖ Fail to reject the Null Hypothesis (not accept the Null Hypothesis)

- ❖ Start with the Alternate Hypothesis. This is what you want to prove.
- ❖ Null Hypothesis will always have the equal to sign ( $=$ ,  $\geq$ ,  $\leq$ ) and Alternate Hypothesis will never have it.

| $H_0$                               | $H_a$                |
|-------------------------------------|----------------------|
| equal ( $=$ )                       | not equal ( $\neq$ ) |
| greater than or equal to ( $\geq$ ) | less than ( $<$ )    |
| less than or equal to ( $\leq$ )    | more than ( $>$ )    |

# NULL AND ALTERNATE HYPOTHESIS

## TIPS

- ❖ Start with the Alternate Hypothesis. This is what you want to prove.
- ❖ Null Hypothesis will always have the equal to sign ( $=, \geq, \leq$ ) and Alternate Hypothesis will never have it.

| $H_0$                               | $H_a$                |
|-------------------------------------|----------------------|
| equal ( $=$ )                       | not equal ( $\neq$ ) |
| greater than or equal to ( $\geq$ ) | less than ( $<$ )    |
| less than or equal to ( $\leq$ )    | more than ( $>$ )    |

# NULL AND ALTERNATE HYPOTHESIS

|            |                                 | True State of Nature |                            |
|------------|---------------------------------|----------------------|----------------------------|
|            |                                 | $H_0$<br>Is true     | $H_a$<br>Is true           |
| Conclusion | Support $H_0$ /<br>Reject $H_a$ | Correct Conclusion   | Type II Error              |
|            | Support $H_a$ /<br>Reject $H_0$ | Type I Error         | Correct Conclusion (Power) |

|                         | Type I error (alpha)                                   | Type II error (beta)                                   |
|-------------------------|--------------------------------------------------------|--------------------------------------------------------|
| Name                    | Producer's risk/<br><b>Significance level</b>          | Consumer's risk                                        |
| 1 minus error is called | <b>Confidence level</b>                                | Power of the test                                      |
| Example of Fire Alarm   | False fire alarm leading to inconvenience              | Missed fire leading to disaster                        |
| Effects on process      | Unnecessary cost increase due to frequent changes      | Defects may be produced                                |
| Control method          | Usually fixed at a pre-determined level, 1%, 5% or 10% | Usually controlled to < 10% by appropriate sample size |
| Simple definition       | Innocent declared as guilty                            | Guilty declared as innocent                            |

# *Hypothesis Testing - Errors*



# P - Value

- ❖ p value is the lowest value of alpha for which the null hypothesis can be rejected. (Probability that the null hypothesis is correct)
- ❖ For example, if  $p = 0.045$  you can reject the null hypothesis at  $\alpha = 0.05$
- ❖ p is low the null must go / p is high the null fly.

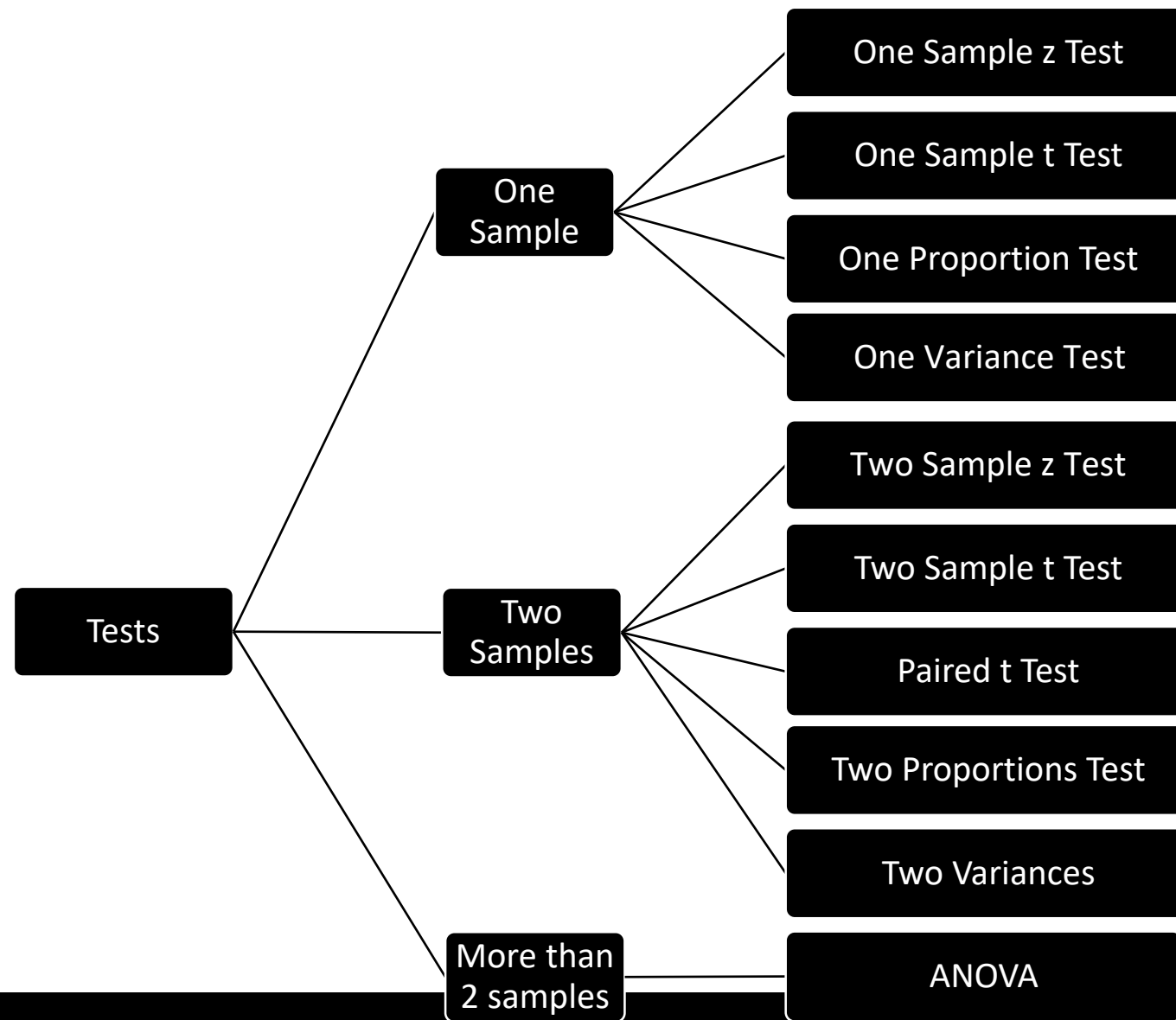
**P - VALUE**



## 5 Hypothesis Tests



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# Hypothesis Tests

1. State the Alternate Hypothesis.
2. State the Null Hypothesis.
3. Select a probability of error level (alpha level). Generally 0.05
4. Calculate the test statistic (e.g t or z score)
5. Critical test statistic
6. Interpret the results.

## Hypothesis Testing - Six Steps

$$z = \frac{(\bar{x} - \mu)}{\sigma / \sqrt{n}}$$

**One Sample z Test**

$$t = \frac{(\bar{x} - \mu)}{s / \sqrt{n}}$$

**One Sample t Test**

$$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

**One Proportion Test**

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

**One Variance Test**

**One Sample Tests – Calculated Values**

$$Z_{cal} = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

**Two Sample z Test**

$$F_{cal} = \frac{s_1^2}{s_2^2}$$

**Two Variance Test**

**Two Sample Tests – Calculated Values**

$$t_{cal} = \frac{(\bar{x} - \mu)}{s/\sqrt{n}}$$

# Two Sample t Tests

❖ Is variance for two populations equal?

Yes

$$t_{cal} = \frac{(\bar{x}_1 - \bar{x}_2)}{s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$df = n_1 + n_2 - 2$$

No

$$t_{cal} = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

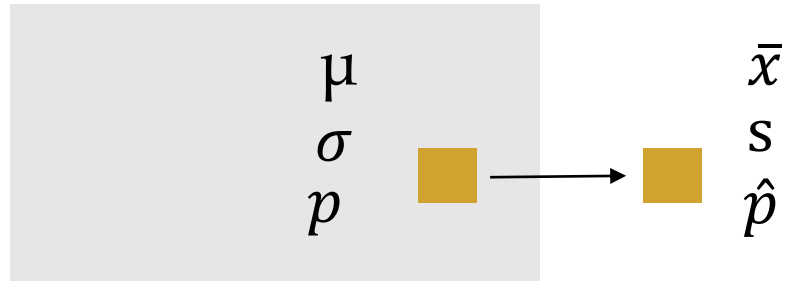
$$df = \frac{\left[\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right]^2}{\frac{(s_1^2/n_1)^2}{(n_1 - 1)} + \frac{(s_2^2/n_2)^2}{(n_2 - 1)}}$$

The approximate degrees of freedom are rounded down to the nearest integer.

## *Two Sample t Test*

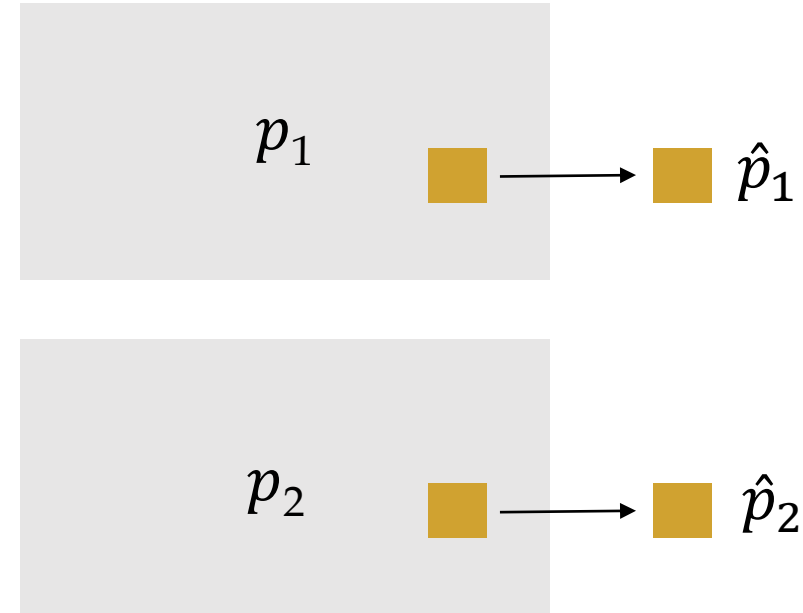
# Proportions - Sample vs Population

How do we represent sample and population proportions?



One Proportion Test

$$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$



## *Two Proportions Test*



## One Proportion Test

$$z = \frac{p - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

# Two Proportions Tests

Test for no difference between proportions

$$H_0: p_1 - p_2 = 0$$

$$H_a: p_1 - p_2 \neq 0$$

Yes

Pooled

No

Un-pooled

$$H_0: p_1 - p_2 = d$$

$$H_a: p_1 - p_2 \neq d$$

$$\bar{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{(n_1 + n_2)}$$

$$z_{cal} = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\bar{p}(1 - \bar{p}) \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$z_{cal} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}}$$

## *Two Proportions Test*

# Variance Tests

- ❖ Chi-square test
  - ❖ For testing the population variance against a specified value
  - ❖ testing goodness of fit of some probability distribution
  - ❖ testing for independence of two attributes (Contingency Tables)
- ❖ F-test
  - ❖ for testing equality of *two* variances from different populations
  - ❖ for testing equality of several means with technique of ANOVA.

***Variance Tests***



## 6 ANOVA and other topics



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### Degrees of freedom

$$\text{Total df} = \text{df}_{\text{factor}} + \text{df}_{\text{error}}$$

### Sum of Square

$$\text{SST} = \text{SS}_{\text{factor}} + \text{SS}_{\text{error}}$$

### Mean Sum of Square = SS / df

$$\text{MSS}_{\text{factor}} = \text{SS}_{\text{factor}} / \text{df}_{\text{factor}}$$

$$\text{MSS}_{\text{error}} = \text{SS}_{\text{error}} / \text{df}_{\text{error}}$$

### F Value

$$F = \text{MSS}_{\text{factor}} / \text{MSS}_{\text{error}}$$

# ***ANOVA***

# Goodness-of-Fit Test

$H_0$ : The data follow a specified distribution.

$H_a$ : The data do not follow the specified distribution.

$$\chi^2 = \sum_{i=1}^k \frac{(O - E)^2}{E}$$

***Goodness of Fit Test***

# Contingency Tables

$H_0$ : The two criteria are independent.

$H_A$ : Some dependence exists.

$$\chi^2 = \sum_{i=1}^k \frac{(O - E)^2}{E}$$

***Contingency Tables***